

SGA 2: Application to the Fundamental Group

Bounded source-aligned working unit — 20 July 2026

Authority. Corrected French TeX line 3456 only; original printed pp. 117–118, physical source-PDF pp. 101–102, and recomposed running pp. 93–94. These locator systems are kept distinct. Blank line 3457 is excluded. The raw continuation cursor is line 3457 and the next substantive cursor is line 3458, the opening of Section 3. The external English candidate contains the complete deferred note and is comparison-only. The French authority remains unchanged.

Editor’s note (3)

Combined with Proposition 3.3 and criteria XII.2.4 and XII.3.4, this yields the following relative Lefschetz theorem. Let $f: X \rightarrow S$ be a projective and flat morphism of connected noetherian schemes, and let D be an effective relative Cartier divisor on X that is relatively ample. If, for every $s \in S$, the depth of X_s at every closed point is ≥ 2 , then D is connected and, for every open subset U of X containing D , the map

$$i_U: \pi_1(D) \longrightarrow \pi_1(U)$$

is surjective. If, in addition, the depth of X_s at every closed point of D_s is ≥ 3 and the local rings of X at its closed points are pure—for example, are complete intersections (cf. X, 3.4)—then i_X is an isomorphism.

Cf. J.-B. Bost, “Lefschetz theorem for Arithmetic Surfaces,” *Ann. Sci. Éc. Norm. Sup. (4)* **32** (1999), pp. 241–312, Theorems 1.1 and 2.1.

When X is merely a smooth projective surface that is geometrically connected over a field, D is always connected and

$$\pi_1(D) \longrightarrow \pi_1(U)$$

is surjective (where U is an open subset containing D), provided only that D is nef and has self-intersection > 0 ; cf. *loc. cit.*, Theorem 2.3, and also Theorem 2.4 for surfaces assumed only normal and complete.

In the case of an arithmetic surface (normal and quasi-projective) X over a ring of integers $\mathcal{O}_K$, Bost, improving results of Y. Ihara (Y. Ihara, “Horizontal divisors on arithmetic surfaces associated with Belyi uniformizations,” in *The Grothendieck theory of dessins d’enfants (Luminy, 1993)*, London Math. Soc. Lect. Note Series, vol. 200, Cambridge Univ. Press, Cambridge, 1994, pp. 245–254, or *loc. cit.*, Corollary 7.2), proved that if a point $P \in X(\mathcal{O}_K)$, which plays the role of the divisor D in the geometric situation, satisfies certain positivity conditions, then the map

$$\pi_1(X) \longrightarrow \pi_1(\mathrm{Spec} \mathcal{O}_K)$$

induced by the projection is an isomorphism whose inverse is the map

$$\pi_1(\mathrm{Spec} \mathcal{O}_K) \longrightarrow \pi_1(X)$$

induced by P ; cf. *loc. cit.*, Theorem 1.2.

Accepted source-normalization note (SGA2-X-L3456-AU-DESSUS-NORMALIZATION-001@1). The corrected French control selects au-dessus from `\sisi{au dessus}{au-dessus}`; the target renders this as “over.” This is an explicit source normalization, not a source defect.